TEAM GRYFFRINDOR

SLEEPWELL ANALYTICS

SLEEP HEALTH & LIFESTYLE DATASET

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OUR TEAM



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ABOUT US

“Better Sleep, Better Life”

Sleep shapes the way we live, yet most of us barely notice its influence. Our daily routines, habits, and environments all leave subtle marks on how we rest, recover, and perform. This project explores the hidden connections between lifestyle and sleep, uncovering the patterns that quietly define our health and well-being.



HYPOTHESIS

01

INDIVIDUAL FACTORS

- Obesity increases likelihood of sleep apnea.
- Higher stress correlates with insomnia.
- Sleeping <6 hours increases disorder risk.
- Low physical activity (<40 min/day) increases disorder prevalence.
- High heart rate / BP increases apnea risk.

02

AGE EFFECTS

- Older individuals are more likely to develop a sleep disorder.
- Age is more strongly associated with sleep apnea than with insomnia.

03

INTEGRATED MODEL

- Multiple risk factors demonstrate cumulative effects on sleep disorder
- Different risk factor combinations predict different disorder types:

BMI + BP + Older age → Sleep Apnea
Stress + Younger age → Insomnia

WHICH ATTRIBUTES CONTRIBUTE TO SLEEP DISORDERS?

LOGISTIC REGRESSION

Dep. variable =
has_disorder → binary

```
Optimization terminated successfully.
         Current function value: 0.450264
         Iterations 7
                           Logit Regression Results
==============================================================================
Dep. Variable:           has_disorder   No. Observations:                  132
Model:                          Logit   Df Residuals:                      124
Method:                           MLE   Df Model:                            7
Date:                Wed, 10 Dec 2025   Pseudo R-squ.:                  0.3451
Time:                        13:17:59   Log-Likelihood:                -59.435
converged:                       True   LL-Null:                       -90.752
Covariance Type:            nonrobust   LLR p-value:                 4.489e-11
============================================================================================
                               coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------------
const                      -25.6230     11.470     -2.234      0.025     -48.104      -3.142
stress_level                -0.1666      0.366     -0.455      0.649      -0.885       0.551
sleep_duration               0.3203      0.794      0.403      0.687      -1.236       1.877
quality_of_sleep            -0.7737      0.586     -1.320      0.187      -1.922       0.375
physical_activity_level     -0.0140      0.025     -0.551      0.582      -0.064       0.036
heart_rate                   0.0483      0.096      0.504      0.614      -0.140       0.236
daily_steps              -1.891e-05      0.000     -0.061      0.951      -0.001       0.001
systolic                     0.2099      0.046      4.550      0.000       0.119       0.300
============================================================================================
```

R-SQUARED: 0.3451 < 1

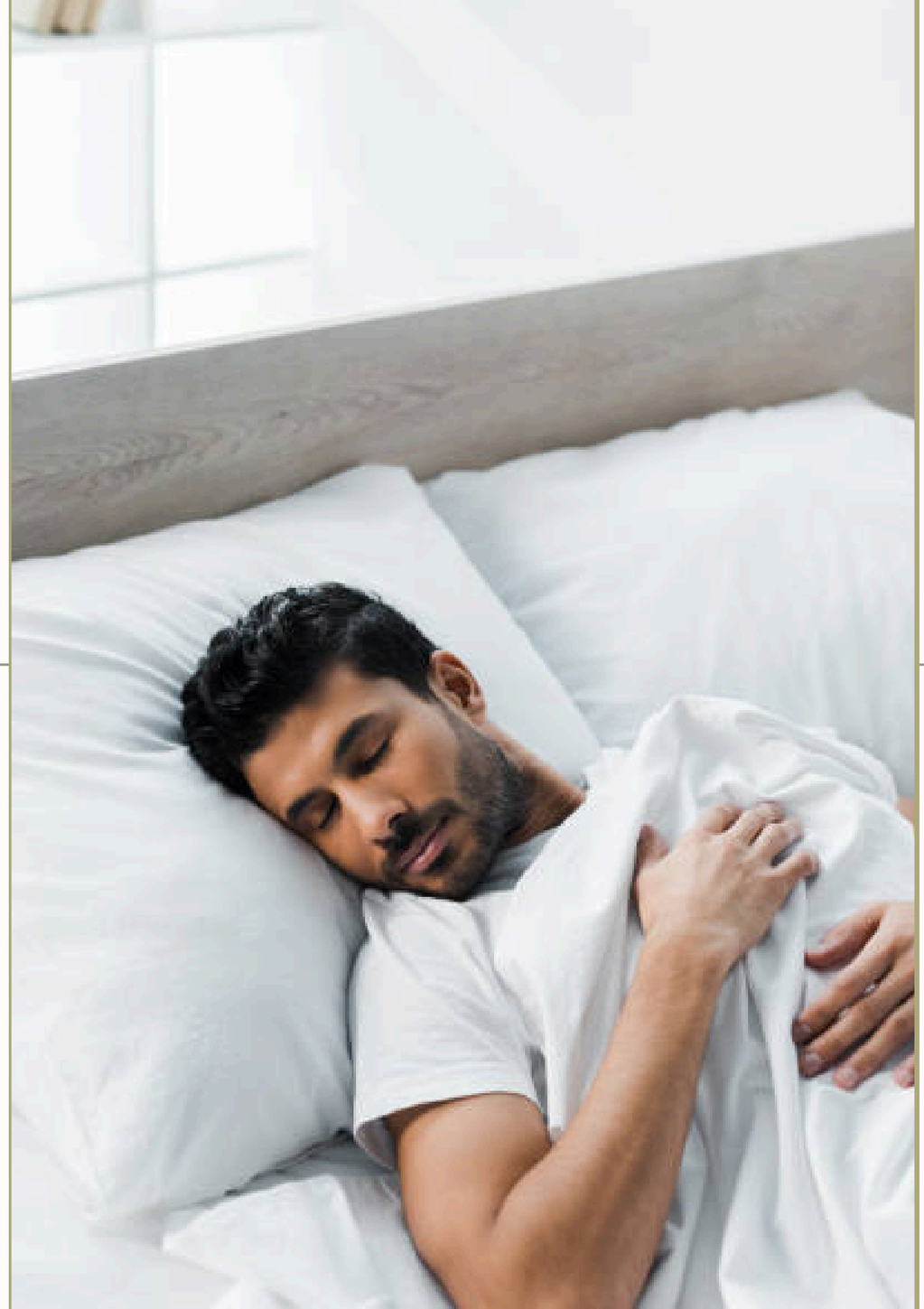
P>|Z| VALUES < 0.05

Only the variable systolic provides strong evidence of which attributes drive sleep disorders.

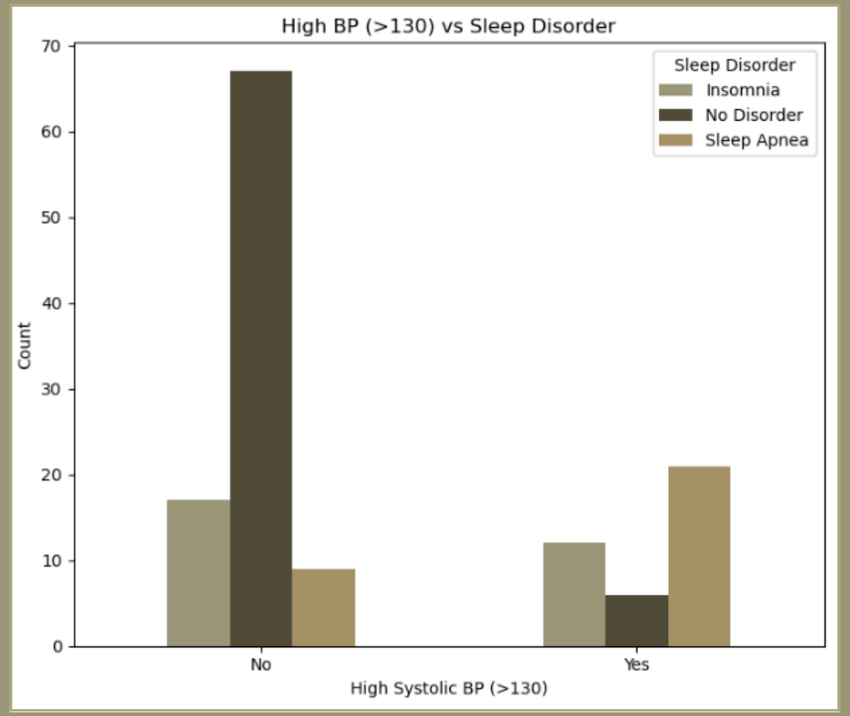
It might be because the sample is too small.

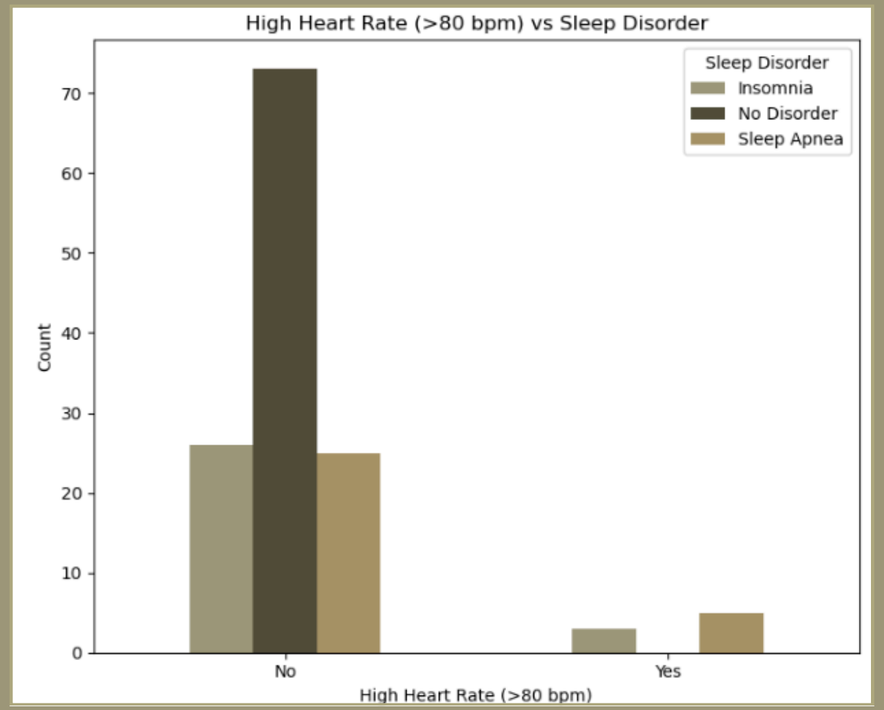
HYPOTHESIS 1

INDIVIDUALS WITH HIGH STRESS, HIGH BMI, LOW SLEEP DURATION, AND POOR SLEEP QUALITY ARE SIGNIFICANTLY MORE LIKELY TO HAVE A SLEEP DISORDER.



DOES BLOOD PRESSURE AFFECT SLEEP ?





WHY IT MATTERS:

- Easy to measure
- Links cardiovascular + sleep
- Already screened routinely

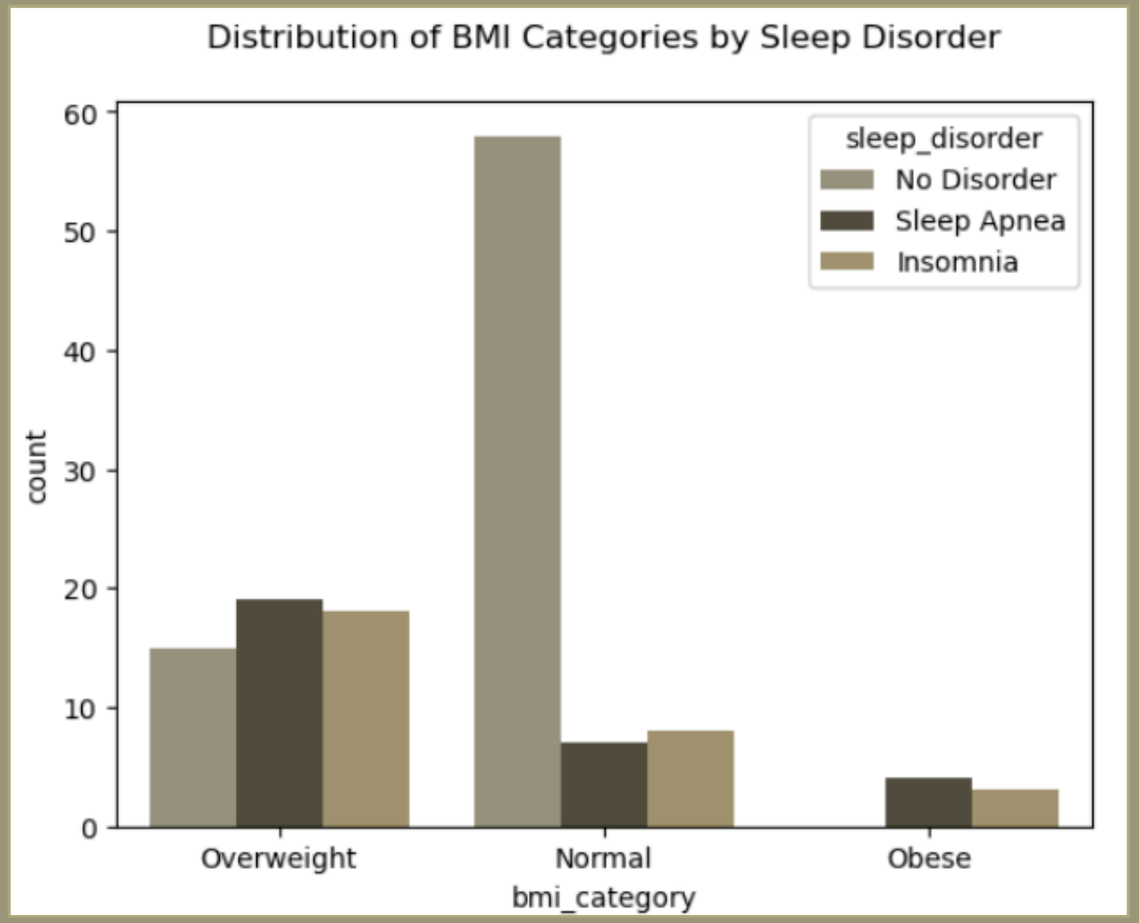
BP > 130:

- 84% disorder rate
- 3x risk elevation

Heart Rate > 80:

- 100% disorder rate
- Small sample (n =8)

DOES WEIGHT AFFECT SLEEP ?



OBESE (n=7)

- 100% disorders
  - 57% Sleep Apnea
  - 43% Insomnia

OVERWEIGHT

- 71% disorders

NORMAL

- 21% disorders

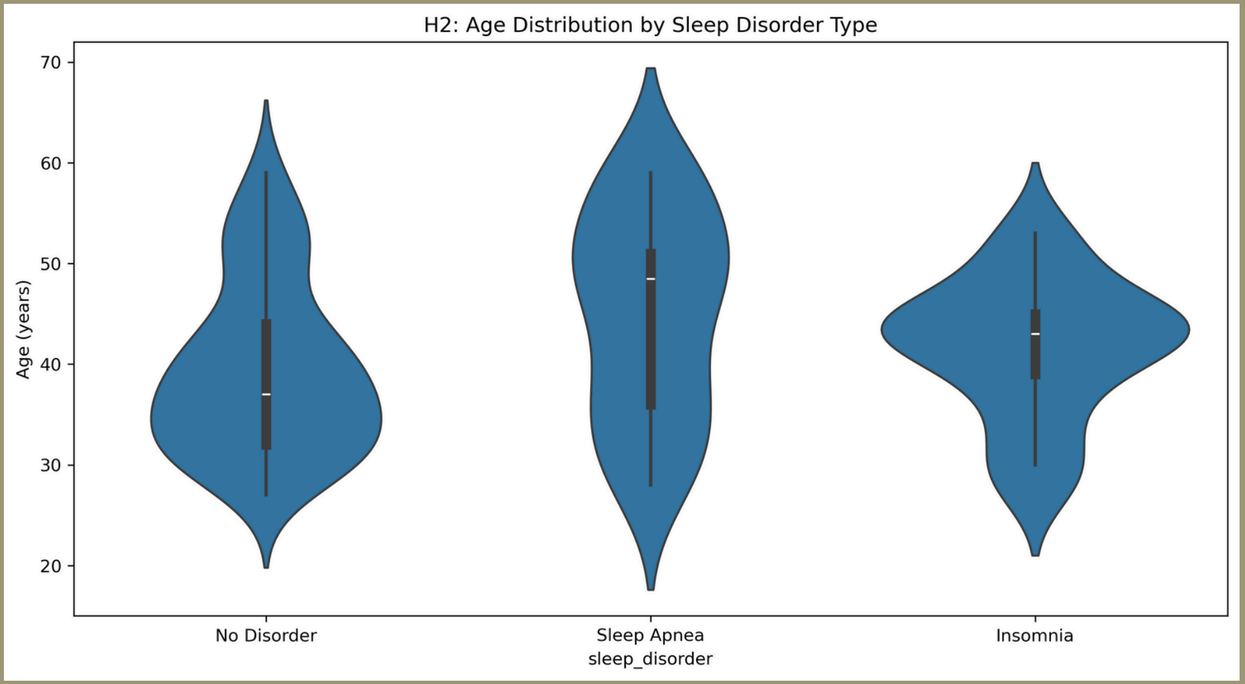
Obesity increases the likelihood of sleep apnea

HYPOTHESIS II

AGE SIGNIFICANTLY INFLUENCES BOTH THE LIKELIHOOD AND THE TYPE OF SLEEP DISORDER.



AGE SHOWS MODERATE INFLUENCE



MEAN AGE

- Sleep Apnea: 44.7
- Insomnia: 41.6
- No Disorder: 39.5

Difference: 5.3 years

SURPRISING FINDING

- Older people → MORE sleep ($r=0.348$, $p<0.001$)

INTERPRETATION:

- Working-age sample (27-59) shows lifestyle effects.
- True age decline may occur post-60.

MODERATE EFFECT - USE WITH OTHER FACTORS

HYPOTHESIS III

MULTIPLE RISK FACTORS DEMONSTRATE CUMULATIVE EFFECTS ON SLEEP DISORDER

INDIVIDUALS WITH MULTIPLE RISK FACTORS (BMI > 30 + BP > 130 + AGE > 45) HAVE SIGNIFICANTLY HIGHER DISORDER PREVALENCE THAN THOSE WITH SINGLE RISK FACTORS



CAN WE COMBINE BMI + BP + AGE + HR TO PREDICT RISK?

Each factor gives you points

WEIGHT

- Normal = 0 pts
- Overweight = +1 pt
- Obese = +3 pts

BLOOD PRESSURE

- Normal ≤130 = 0 pts
- High >130 = +2 pts

AGE

UNDER 45 = 0 PTS
45+ = +1 PT

HEART RATE

NORMAL ≤80 = 0 PTS
HIGH >80 = +1 PT

TWO EXAMPLES:

PERSON A

NORMAL WEIGHT (0 PTS)
✓ BP: 120 (0 PTS)
✓ AGE: 35 (0 PTS)
✓ HR: 70 (0 PTS)

0 POINTS
LOW RISK 🟢

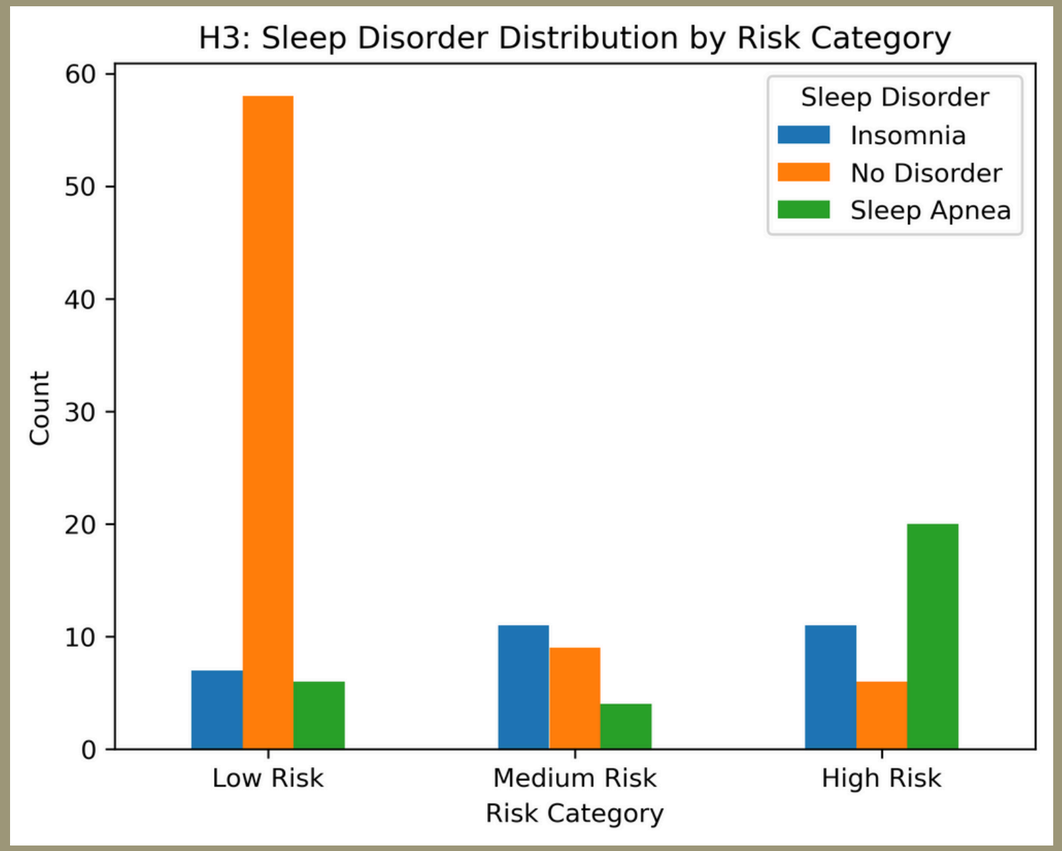
PERSON B

⚠ OBESE (+3 PTS)
⚠ BP: 145 (+2 PTS)
⚠ AGE: 52 (+1 PT)
⚠ HR: 85 (+1 PT)

7 POINTS
HIGH RISK 🔴

DOES THE RISK SCORE PREDICT SLEEP DISORDERS?

DISORDER PREVALENCE BY RISK CATEGORY



Low Risk → Majority have no disorder
Medium Risk → Insomnia becomes more prevalent
High Risk → Sleep Apnea becomes dominant

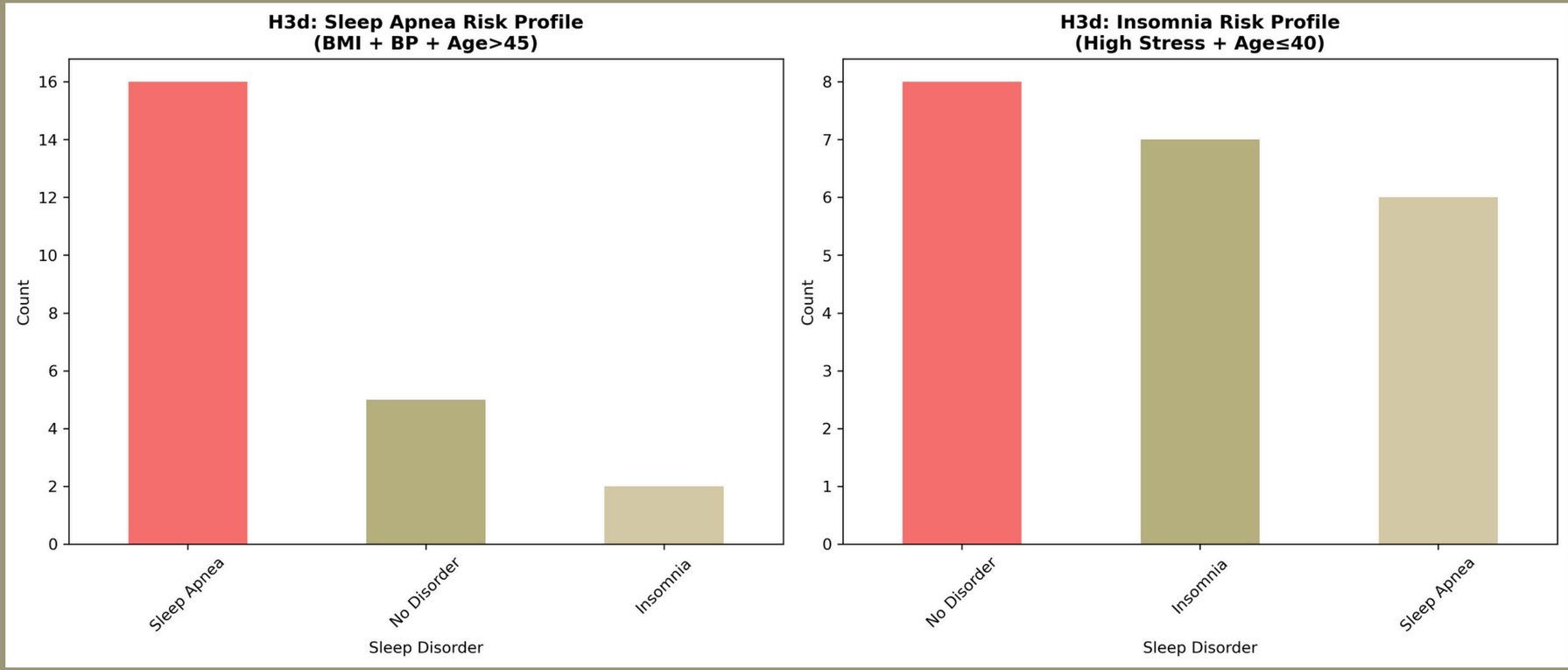
Risk Score Categories correspond strongly to real sleep disorder outcomes

RISK LEVEL	% POPULATION	DISORDER RATE
LOW (0-1 pts)	55%	18%
MEDIUM (2-3 pts)	17%	62%
HIGH (4+ pts)	28%	84%

Key Finding: Higher risk score = higher disorder rate (18% ⇢ 62% ⇢ 84%).

CAN WE PREDICT DISORDER TYPE?

APNEA VS INSOMNIA PROFILE



SLEEP APNEA PROFILE

- BMI + BP + Age>45
- 70% Sleep Apnea
- 3x risk elevation
- Ready for deployment

INSOMNIA PROFILE

- High Stress + Young Age
- 33% Insomnia
- 1.5x risk elevation
- Needs behavioral data

TAKEAWAY:

Physiological disorders predictable,
psychological disorders need more variables

OUR FINDINGS

VALIDATED

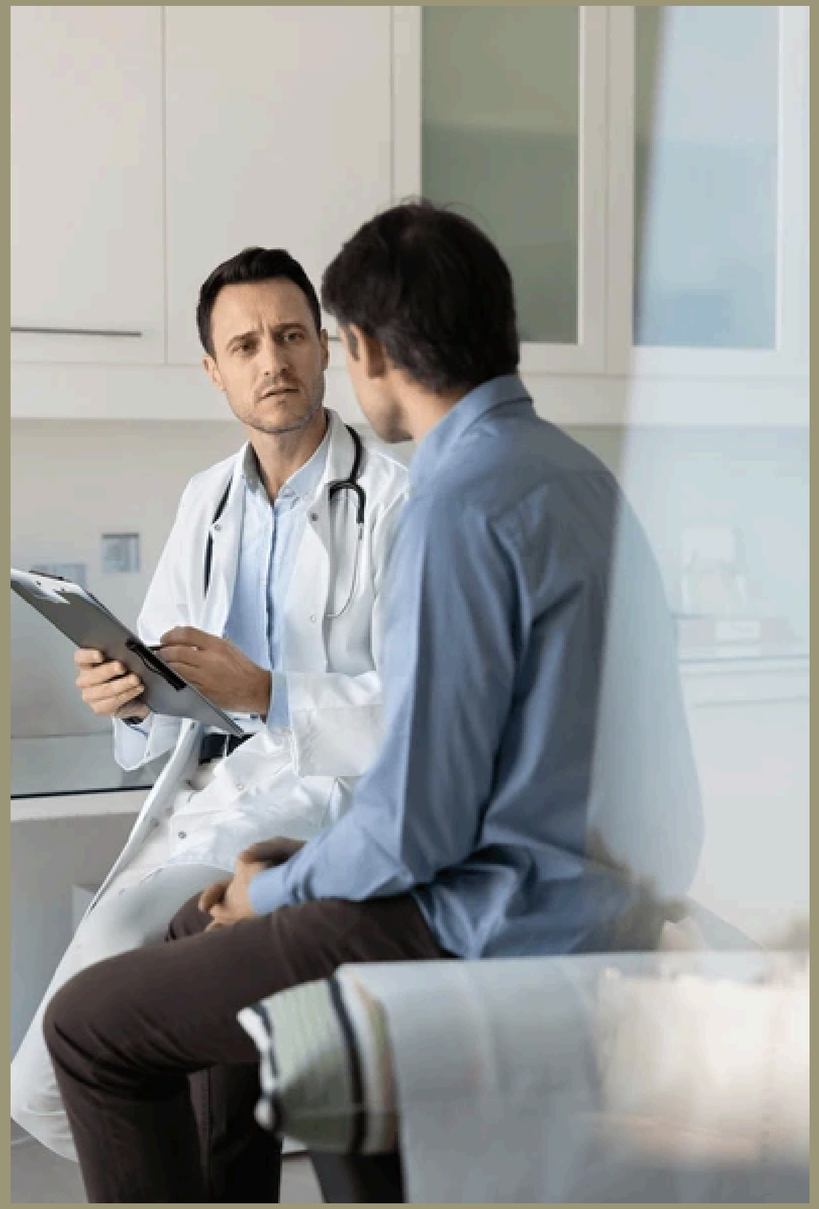
- Cardiovascular markers (BP, HR) = strongest predictors.
- BMI/Obesity = 100% disorder rate (6x apnea risk).
- Combined risk model = 84% detection in high-risk
- Simple 4-parameter tool = clinically feasible

NOT SUPPORTED

- Stress alone insufficient for insomnia prediction.
- Physical activity showed no relationship.
- Sleep duration <6h (insufficient data, n=4).

LIMITATIONS

- Small sample size (n=132).
- Age range limited to working-age (27-59).



WHAT SHOULD HEALTH CARE DO?

IMMEDIATE (0-3 MONTHS)

- DEPLOY RISK CALCULATOR IN PRIMARY CARE
- PILOT IN 2-3 CLINICAL SITES
- TRAIN PHYSICIANS ON SCORING SYSTEM
- PRIORITIZE HIGH-RISK FOR SLEEP STUDIES

VALIDATE (3-12 MONTHS)

- TEST ON LARGER COHORT (N=500+)
- EXPAND AGE RANGE TO INCLUDE 60+
- TRACK REAL-WORLD ACCURACY
- REFINE INSOMNIA PREDICTION

SCALE (12+ MONTHS)

- INTEGRATE WITH ELECTRONIC HEALTH RECORDS
- POPULATION HEALTH PROGRAMS
- EMPLOYER WELLNESS INITIATIVES
- SHIFT FROM REACTIVE TO PREVENTIVE CARE

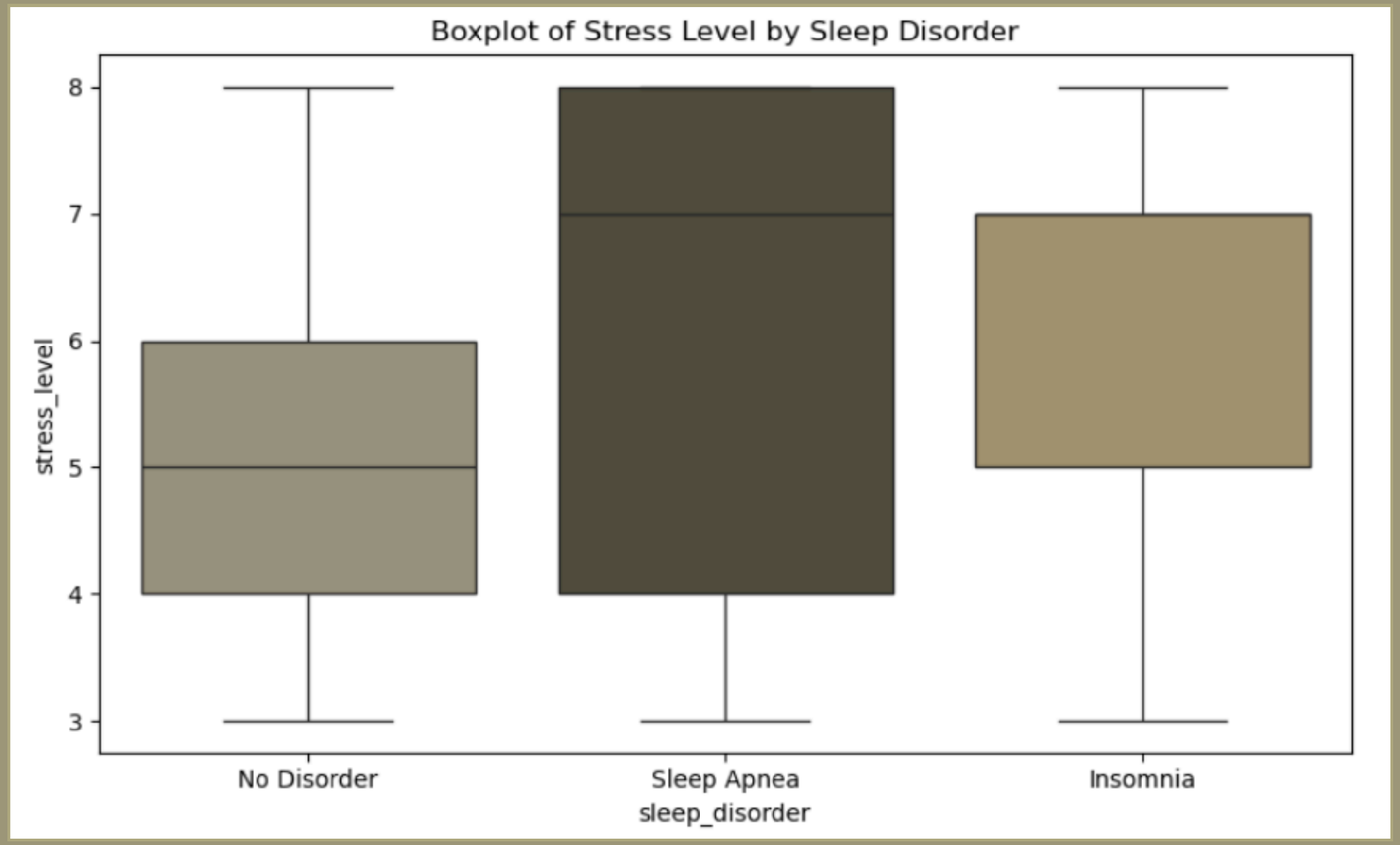


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THANK YOU FOR WATCHING!

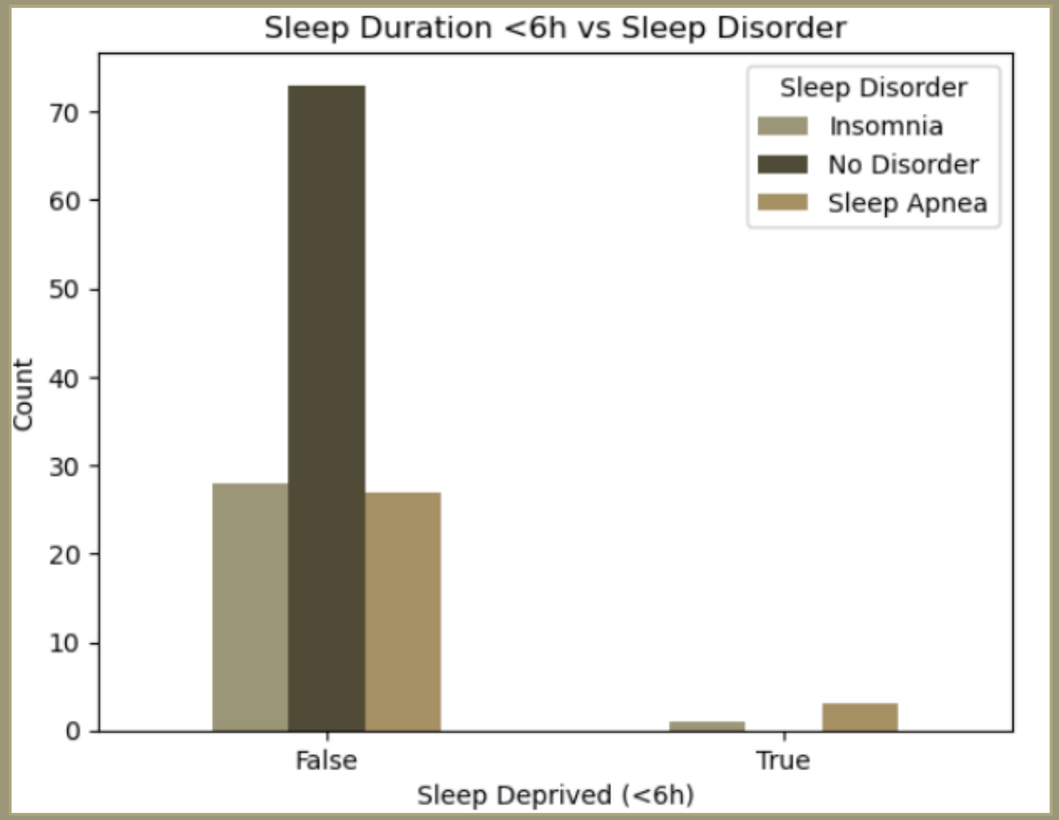
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HIGHER STRESS CORRELATES WITH INSOMNIA



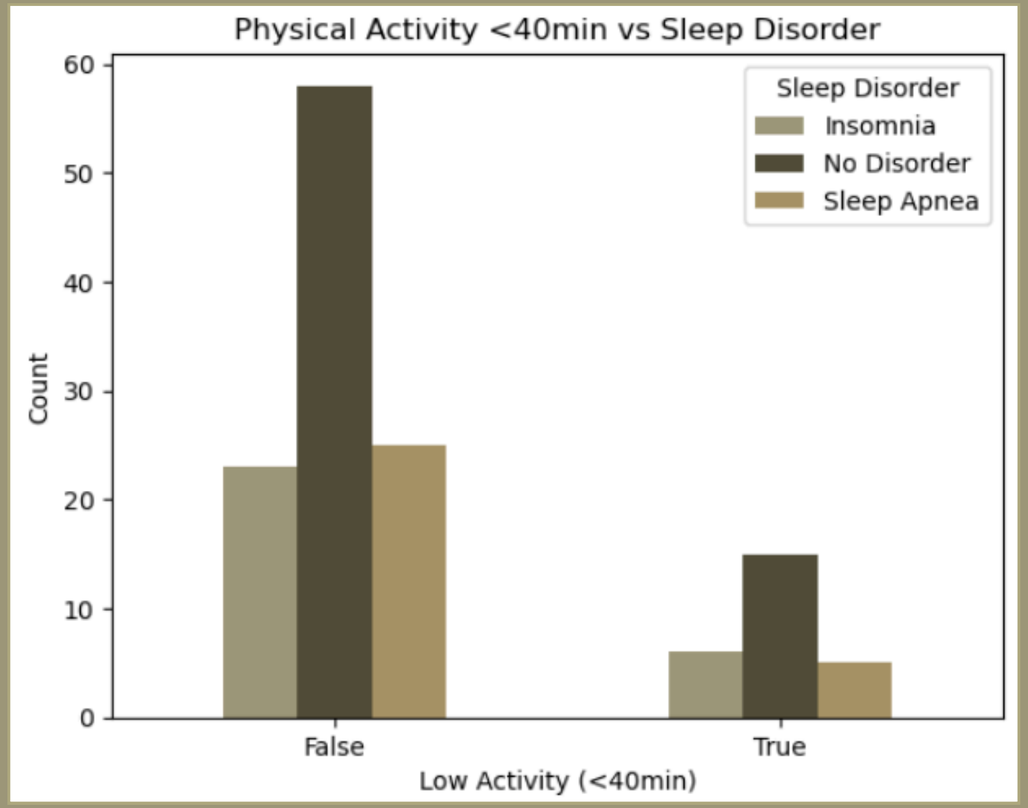
The main finding is that **higher stress correlates with insomnia.**

SLEEPING <6 HOURS INCREASES DISORDER RISK



This point supports the general Hypothesis I, which includes **low sleep duration** as a factor making individuals **more likely to have a sleep disorder.**

LOW PHYSICAL ACTIVITY (<40 MIN/DAY) INCREASES DISORDER PREVALENCE.



The primary conclusion is that **low physical activity** (less than 40 minutes per day) **increases the prevalence of sleep disorders.**

- Low physical activity is explicitly defined as <40 min/day in the document's hypothesis section.
- This finding aligns with Hypothesis I, suggesting that lifestyle factors like activity level are connected to sleep health.

SUMMERIZE

HYPOTHESIS I

SUMMERIZE

